analytical structure of inventory problems, simple models with deterministic and stochastic demand with and without lead time, storage models with particular reference to dam type.

Homogeneous discrete-time Markov chains, transition probability matrix, classification of states and ergodic theorems, homogeneous continuous-time Markov chains, Poisson process, elements of queuing theory, M/M/1, M/M/K, G/M/1 and M/G/1 queues.

Solution of statistical problems on computers using well-known statistical software packages like SPSS.

3. Quantitative Economics and Official Statistics—Determination of trend, seasonal and cyclical components, Box-Jenkins method, tests for stationary series, ARIMA models and determination of orders of autoregressive and moving average components, forecasting.

Commonly used index numbers-Laspeyre's, Paasche's and Fisher's ideal index numbers, chain-base index number, uses and limitations of index numbers, index number of wholesale prices, consumer prices, agricultural production and industrial production, test for index numbers - proportionality, time-reversal, factor-reversal and circular .

General linear model, ordinary least square and generalized least squares methods of estimation, problem of multicollinearity, consequences and solutions of multicollinearity, autocorrelation and its consequences, heteroscedasticity of disturbances and its testing, test for independence of disturbances, concept of structure and model for simultaneous equations, problem of identification-rank and order conditions of identifiability, two-stage least square method of estimation.

Present official statistical system in India relating to population, agriculture, industrial production, trade and prices, methods of collection of official statistics, their reliability and limitations, principal publications containing such statistics, various official agencies responsible for data collection and their main functions.

4. Demography and Psychometry—Demographic data from census, registration, NSS other surveys, their limitations and uses, definition, construction and uses of vital rates and ratios, measures of fertility, reproduction rates, morbidity rate, standardized death rate, complete and abridged life tables, construction of life tables from vital statistics and census returns, uses of life tables, logistic and other population growth curves, fitting a logistic curve, population projection, stable population, quasi-stable population, techniques in estimation of demographic parameters, standard classification by cause of death, health surveys and use of hospital statistics.

Methods of standardisation of scales and tests, Z-scores, standard scores, T-scores, percentile scores, intelligence quotient and its measurement and uses, validity and reliability of test scores and its determination, use of factor analysis and path analysis in psychometry.

ZOOLOGY

PAPER–I

1. Non-chordata and Chordata :

(a) Classification and relationship of various phyla up to subclasses : Acoelomate and Coelomate, Protostomes and Deuterostomes, Bilateria and Radiata; Status of Protista, Parazoa, Onychophora and Hemichordata; Symmetry.

(b) Protozoa : Locomotion, nutrition, reproduction, sex; General features and life history of *Paramaecium, Monocystis, Plasmodium* and *Leishmania*.

(c) Porifera : Skeleton, canal system and reproduction.

(d) Cnidaria : Polymorphism, defensive structures and their mechanism; coral reefs and their formation; metagenesis; general features and life history of *Obelia* and *Aurelia*.

(e) Platyhelminthes : Parasitic adaptation; general features and life history of *Fasciola* and *Taenia* and their pathogenic symptoms.

(f) Nemathelminthes : General features, life history, parasitic adaptation of *Ascaris* and *Wuchereria*.

(g) Annelida : Coelom and metamerism; modes of life in polychaetes; general features and life history of Nereis, earthworm and leach.

(h) Arthropoda : Larval forms and parasitism in Crustacea; vision and respiration in arthropods (Prawn, cockroach and scorpion); modification of mouth parts in insects (cockroach, mosquito, housefly, honey bee and butterfly); metamorphosis in insect and its hormonal regulation, social behaviour of *Apis* and termites.

(i) Mollusca : Feeding, respiration, locomotion, general features and life history of *Lamellidens*, Pila and Sepia, torsion and detorsion in gastropods.

(j) Echinodermata : Feeding, respiration, locomotion, larval forms, general features and life history of *Asterias*.

(k) Protochordata : Origin of chordates; general features and life history of *Branchiostoma* and *Herdmania*.

(l) Pisces : Respiration, locomotion and migration.

(m) Amphibia : Origin of tetrapods, parental care, paedomorphosis.

(n) Reptilia; Origin of reptiles, skull types, status of Sphenodon and crocodiles.

(o) Aves : Origin of birds, flight adaptation, migration.

(p) Mammalia : Origin of mammals, dentition, general features of egg laying mammals, pouched-mammals, aquatic mammals and primates, endocrine glands (pituitary, thyroid, parathyroid, adrenal, pancreas, gonads) and their interrelationships.